



Move		
ACTION	KEY	DESCRIPTION
(MOVE_L / MOVE_R) tap ≤1/5s	< / > tap	Locked cruise at SPEED
(MOVE_L / MOVE_R) hold >1/5s	< / > hold	Hold-to-run; stop on release
Same MOVE tip while locked	< or > tip	Stop cruise
Opposite MOVE	< or >	Reverse immediately
OPTION + (MOVE_L / MOVE_R) hold	* (< / >) hold	Max speed + max accel boost
(FAST_L / FAST_R) hold	<< / >> hold	Max speed/accel jog while held
Matching FAST while cruise	<< or >> hold	Boost to max; release → pots
MOVE_L + MOVE_R hold ≥3s	< > hold ≥3s	Swap L/R
FAST_L + FAST_R hold ≥3s	<< >> hold ≥3s	Swap L/R
STOP tap	0 tap	Smooth stop
STOP tap while slowing	0 tap	Fast halt
STOP hold ≥1s	0 hold ≥1s	Fast halt
OPTION + STOP (both *)	* 0 *	Emergency halt
STOP + A	0 A	Goto soft min
STOP + B	0 B	Goto midpoint
STOP + C	0 C	Goto soft max
OPTION + STOP + A	* 0 A	Homing
DELAY hold while moving	D hold	Soft-pause
DELAY release while paused	D release	Resume

Tap ≤ JKS_MOVE_TAP_MS (333 ms) locks cruise; longer hold stops on release. Same MOVE tip while locked stops.

A / B / C		
ACTION	KEY	DESCRIPTION
(A / B / C) tap	(A / B / C) tap	Goto at SPEED
(A / B / C) hold ≥1s	(A / B / C) hold ≥1s	Save PosA/B/C
OPTION + (A / B / C) tap	* (A / B / C) tap	Goto at max speed/accel
(A + B) / (A + C) / (B + C)	(A B) / (A C) / (B C)	Start/stop loop (2nd letter first)
OPTION + (A + B) / ...	* + loop	Same loop (starts at 2nd letter)
A + B + C hold ≥1s	A B C hold ≥1s	Reset marks
OPTION + STOP (×1 / continuous)	* 0	Peek PosA/B/C on OLED

Display		
ACTION	KEY	DESCRIPTION
OPTION + FAST_L + FAST_R hold ≥1s	* << >> hold ≥1s	Dim LED + OLED (full ↔ 25%)

Other		
ACTION	KEY	DESCRIPTION
STOP hold ≥2s	0 hold ≥2s	Disable motor driver
STOP tap when disabled	0 tap	Enable driver
OPTION + MOVE_L + MOVE_R hold ≥1s	* < > hold ≥1s	Swap L/R for MOVE, FAST, joystick

Pots	
CONTROL	DESCRIPTION
SPEED	Left → 0; right → faster (finer at low end). ~0 → Set SPEED, won't start
ACCEL	How quickly the slider speeds up / slows down

OLED mark ETAs (->B ... s) follow both pots — dial SPEED to match a planned move time.

Joystick		
ACTION	KEY	DESCRIPTION
Centre	—	Stop
Deflect	—	Move; full = SPEED pot
While deflected	—	Overrides MOVE / FAST / loop
OPTION + joystick hold	* hold	Full deflection = max speed
OPTION + A + B + C hold ≥1s	* A B C hold ≥1s	Calibrate centre (stopped)
MOVE_L + MOVE_R hold ≥3s	< > hold ≥3s	Swap L/R incl. joystick
FAST_L + FAST_R hold ≥3s	<< >> hold ≥3s	Swap L/R incl. joystick

Delay		
ACTION	KEY	DESCRIPTION
DELAY tap	D tap	Delay off
DELAY hold N s, release	D hold N s, release	Arm walk-in delay
DELAY hold	D hold	Preview while arming
OPTION + DELAY tap	* D tap	Arm preset (~5 s)
OPTION + DELAY hold	* D hold	Arm time ×5
STOP tap (during wait)	0 tap	Cancel wait (keeps Delay)

Timelapse		
ACTION	KEY	DESCRIPTION
TIMELAPSE tap	T tap	Cycle divider 1...100
TIMELAPSE hold ≥1s	T hold ≥1s	Back to ×1
OPTION + TIMELAPSE tap	* T tap	Divider +1
OPTION + TIMELAPSE hold ≥1s	* T hold ≥1s	Favourite (often ×25)
OPTION + DELAY + TIMELAPSE	* D T	Toggle MSM ↔ continuous
OPTION + STOP (MSM, TL≠1)	* 0	Cycle camera FPS

TL×1 & continuous: CTRL_CAMERA high while moving / soft-paused; low when idle. OLED: **MSM xN @Ffps / Cont xN @Ffps**.

- Timelapse workflows**
 - Plan the path
 - Frame: store PosA / PosB (PosC if needed)
 - Rehearse A→B at TL×1; dial SPEED/ACCEL until pacing feels right
 - MSM: SPEED = hop speed (interval = N/FPS). Continuous: crawl = SPEED÷N
 - Toggle MSM ↔ continuous (* D T); set TL; shoot
 - MSM
 - Wire GP22 shutter (optocoupler); manual exposure/focus
 - Plan at ×1; optional DELAY arm; DELAY hold soft-pauses (freezes take)
 - MSM + TL; OLED **MSM xN @Ffps**; FPS via OPTION+STOP
 - Start camera stills/REC
 - Shoot: B / MOVE / loop (pulse → hop → settle)
 - Stop REC; TL too fast / Step slow if hop won't fit
 - continuous
 - Same wire (hold-high, not pulses)
 - Plan at ×1; match camera TL to slider TL
 - Continuous + TL; OLED Cont xN @Ffps (OPTION+STOP peeks marks only)
 - Start camera REC before the move
 - Shoot: B / MOVE / loop (crawl ÷N); soft-pause keeps pin high
 - Stop REC when idle / after STOP

Status LED	
COLOUR	MEANING
Rainbow	Power-on / unlock
Dim orange	Driver off
Dim white	Idle, ready
Dim cyan	Delay armed
Cyan blink	Delay wait / countdown
Dim magenta	Timelapse idle (TL ≠ 1)
White ↔ blue	Loop idle / dwell
Off (brief)	MSM shutter pulse
Yellow	Accel / decel
Green	Steady speed
+30% / 100% blue	Near / at soft limit
+10% blue	Loop running
Red fast blink	Hard limit
Red blink	Homing
Red	DRV_ERROR / E-stop
Red flash	Halt confirm

Pico pinout — keypad
UART to SliderMC @ 1Mbaud (GP16/17 crossed). Handshake: UIC \n → MC welcome # ... banner.